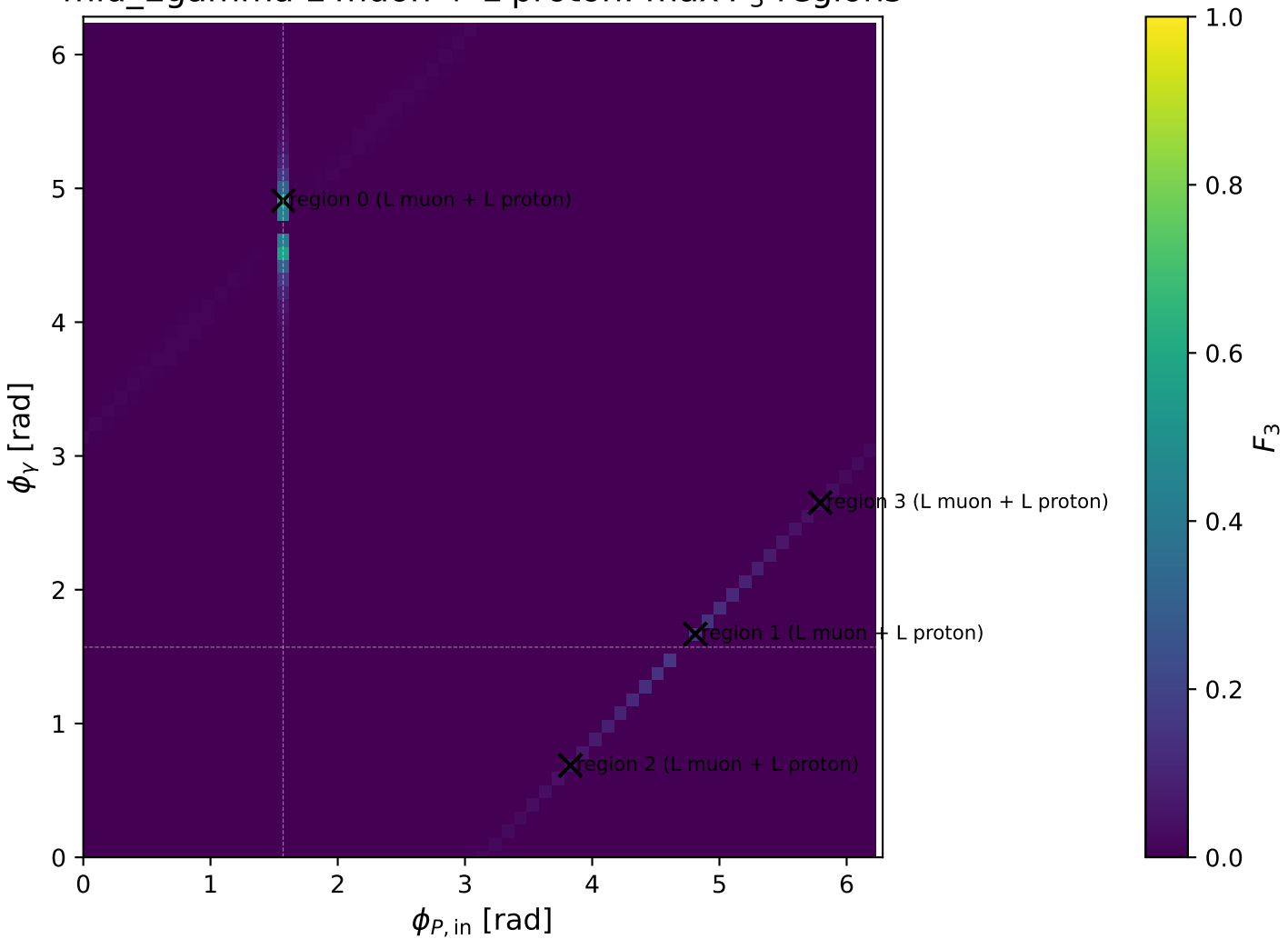
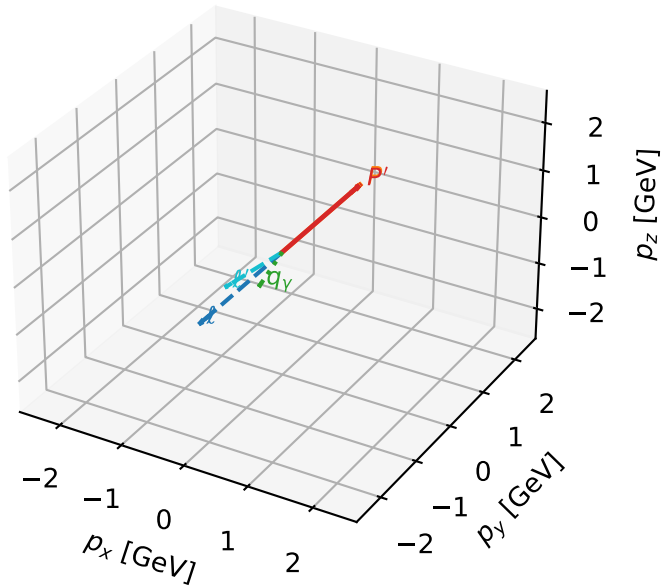


mid_Egamma L muon + L proton: max F_3 regions



L muon + L proton characteristic max F_3 configuration

3D momenta

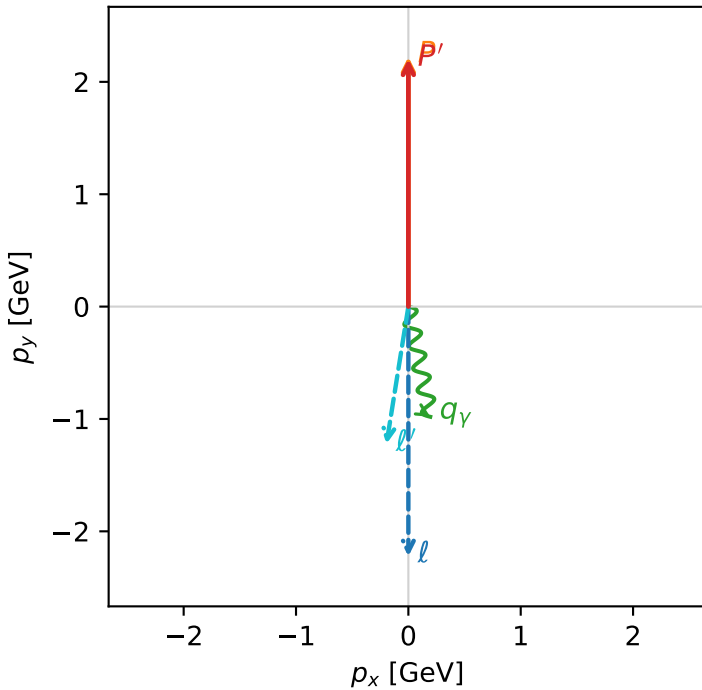


f3_mid_egamma_ll_region_0 F_3=0.578019
region: mid_Egamma_LL_region_0
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_mid_Egamma
incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

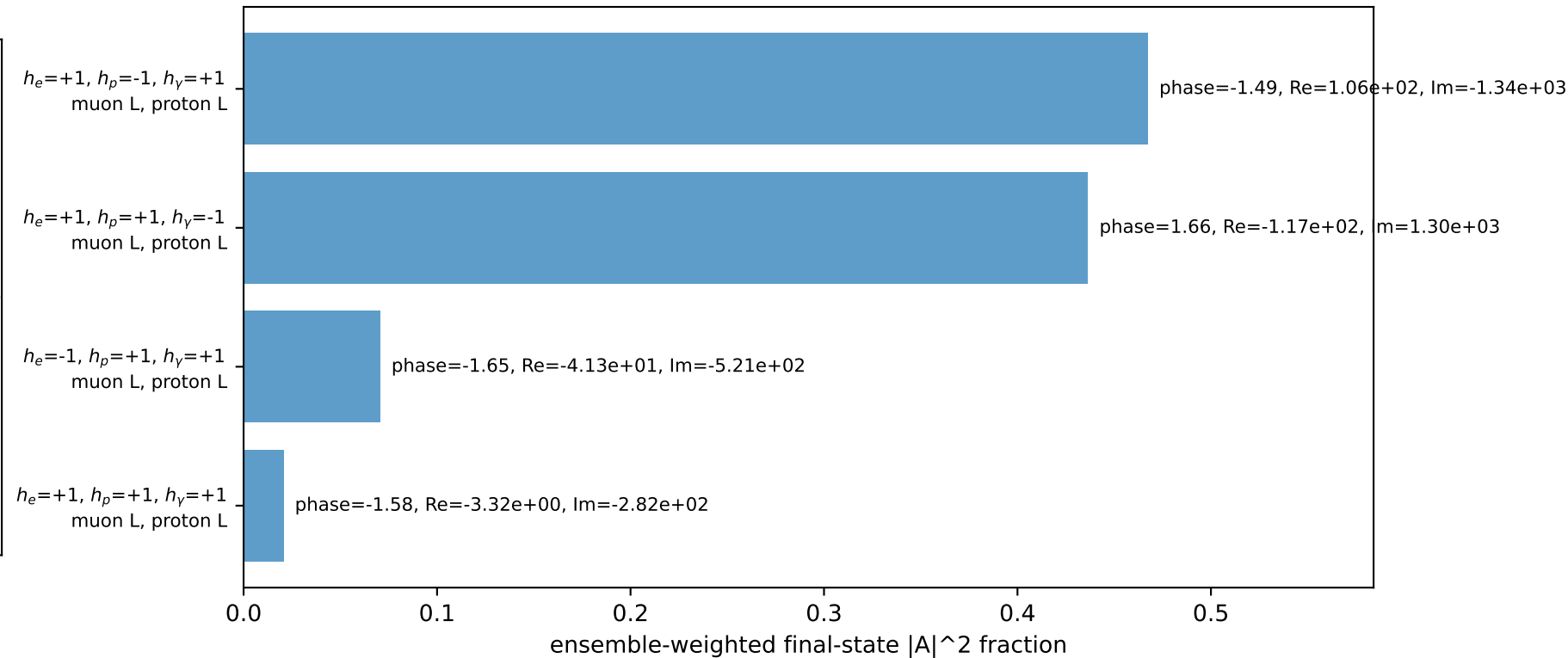
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=2.20402$
 $\theta_{in}=1.57079$, $\phi_l=4.71239$, $\phi_p=1.5708$
 $E_Y=1$, $\phi_Y=4.90874$
 $Q^2=0.0726476$, $x_B=0.00790796$, $t=-5.54648e-05$, $W^2=9.99383$, $y=0.445416$
 $\theta(l', \gamma)=20.3116$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2256$ $p_x= -0.0000$ $p_y= -2.2231$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 0.0000$ $p_y= 2.2231$ $p_z= 0.0000$
 ℓ' $E= 1.2432$ $p_x= -0.1951$ $p_y= -1.2232$ $p_z= 0.0000$
 P' $E= 2.3953$ $p_x= 0.0000$ $p_y= 2.2040$ $p_z= 0.0000$
 q_Y $E= 1.0000$ $p_x= 0.1951$ $p_y= -0.9808$ $p_z= 0.0000$

Transverse plane

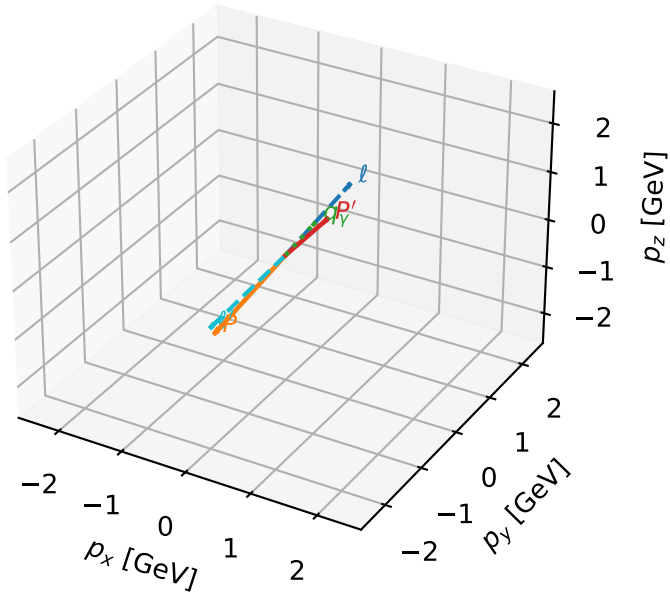


Leading initial-ensemble/final-state components (N=4, retained=99.4%)



L muon + L proton characteristic max F_3 configuration

3D momenta



f3_mid_egamma_ll_region_1 F_3=0.159005

region: mid_Egamma_LL_region_1

outgoing-state purity: 1

kinematic point: high_s_high_theta_in_mid_Egamma

incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.15252$

$\theta_{in}=1.57079$, $\phi_l=1.66897$, $\phi_p=4.81056$

$E_\gamma=1$, $\phi_\gamma=1.66897$

$Q^2=19.105$, $x_B=0.965727$, $t=-10.511$, $W^2=1.55787$, $y=0.959186$

$\theta(l', \gamma)=176.988$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:

l $E= 2.2256$ $p_x= -0.2179$ $p_y= 2.2124$ $p_z= -0.0000$

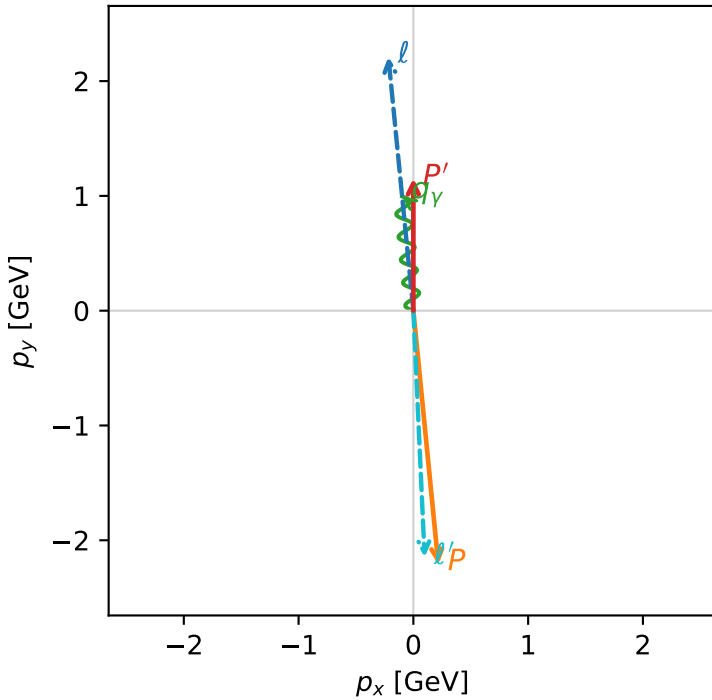
P $E= 2.4129$ $p_x= 0.2179$ $p_y= -2.2124$ $p_z= 0.0000$

l' $E= 2.1525$ $p_x= 0.0980$ $p_y= -2.1477$ $p_z= 0.0000$

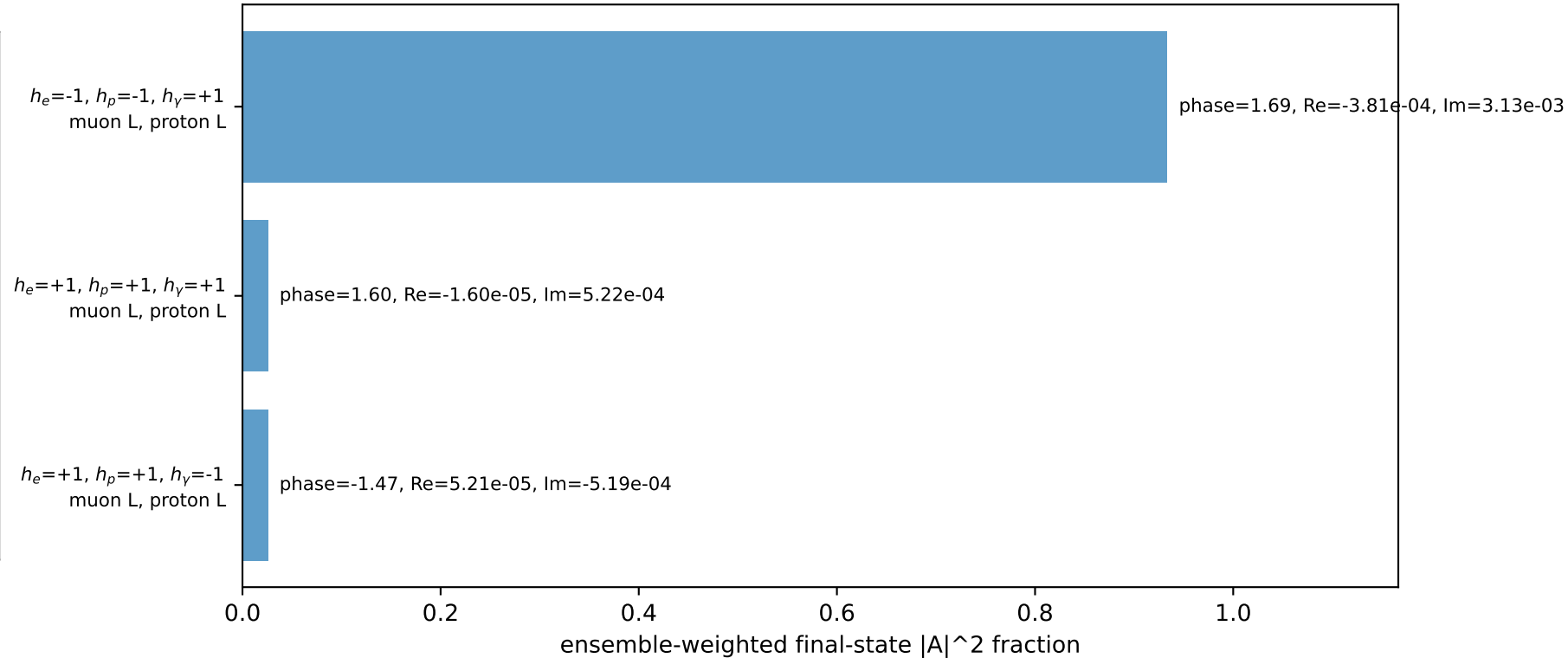
P' $E= 1.4860$ $p_x= 0.0000$ $p_y= 1.1525$ $p_z= 0.0000$

q_γ $E= 1.0000$ $p_x= -0.0980$ $p_y= 0.9952$ $p_z= 0.0000$

Transverse plane

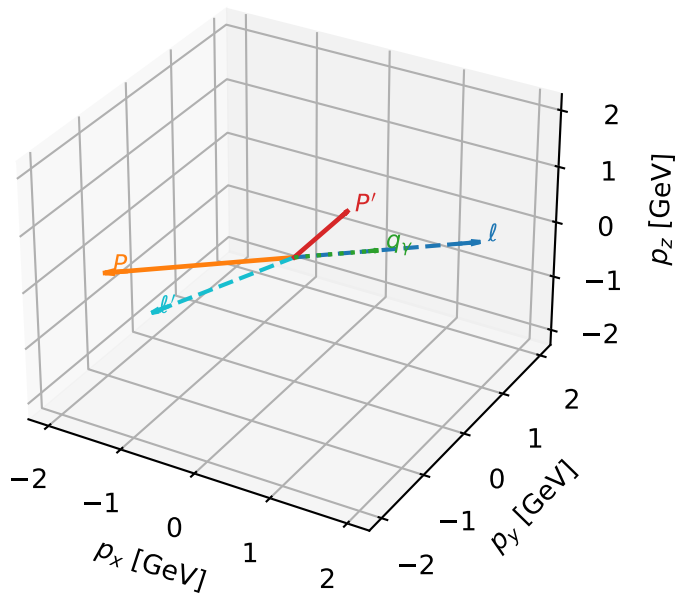


Leading initial-ensemble/final-state components (N=3, retained=98.4%)



L muon + L proton characteristic max F_3 configuration

3D momenta

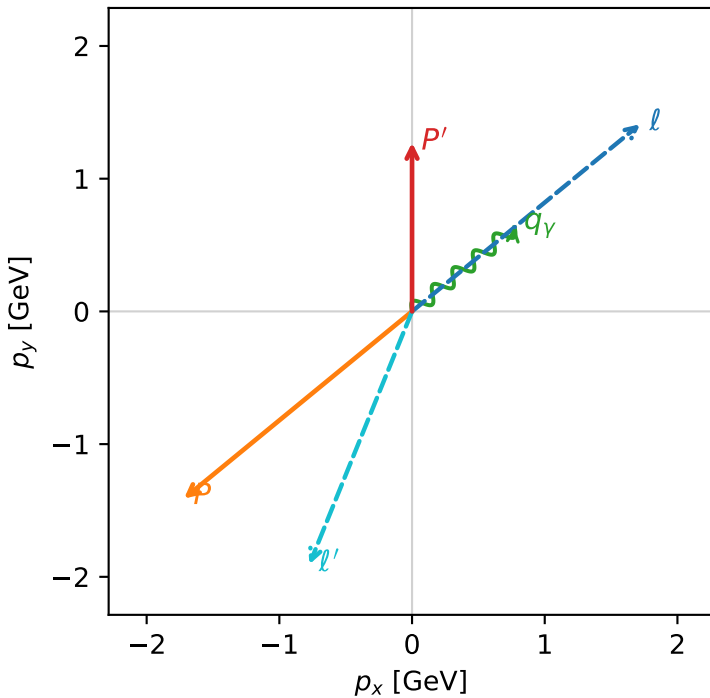


f3_mid_egamma_ll_region_2 F_3=0.0505491
 region: mid_Egamma_LL_region_2
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

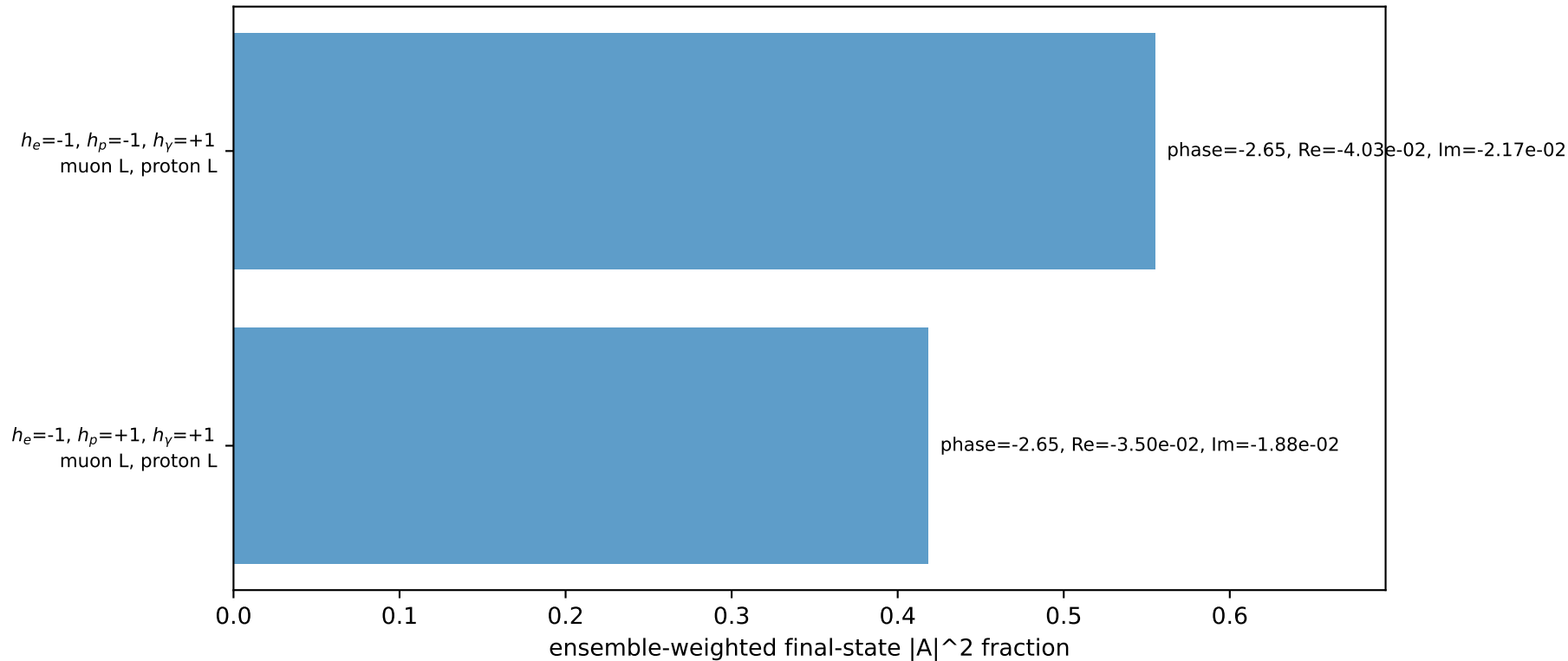
$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.27096$
 $\theta_{in}=1.57079$, $\phi_l=0.687223$, $\phi_p=3.82882$
 $E_\gamma=1$, $\phi_\gamma=0.687223$
 $Q^2=17.1735$, $x_B=0.91738$, $t=-9.44814$, $W^2=2.4265$, $y=0.907651$
 $\theta(l', \gamma)=151.458$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= 1.7185$ $p_y= 1.4103$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -1.7185$ $p_y= -1.4103$ $p_z= 0.0000$
 l' $E= 2.0589$ $p_x= -0.7730$ $p_y= -1.9054$ $p_z= 0.0000$
 P' $E= 1.5796$ $p_x= 0.0000$ $p_y= 1.2710$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= 0.7730$ $p_y= 0.6344$ $p_z= 0.0000$

Transverse plane

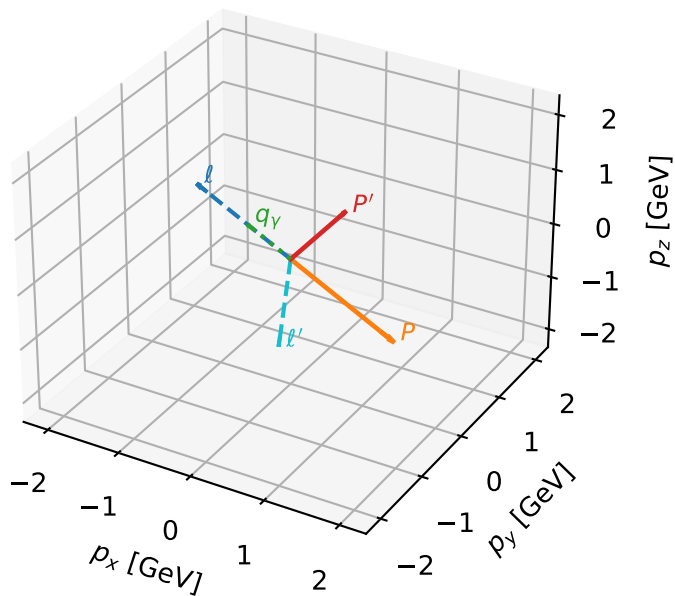


Leading initial-ensemble/final-state components (N=2, retained=97.4%)



L muon + L proton characteristic max F_3 configuration

3D momenta

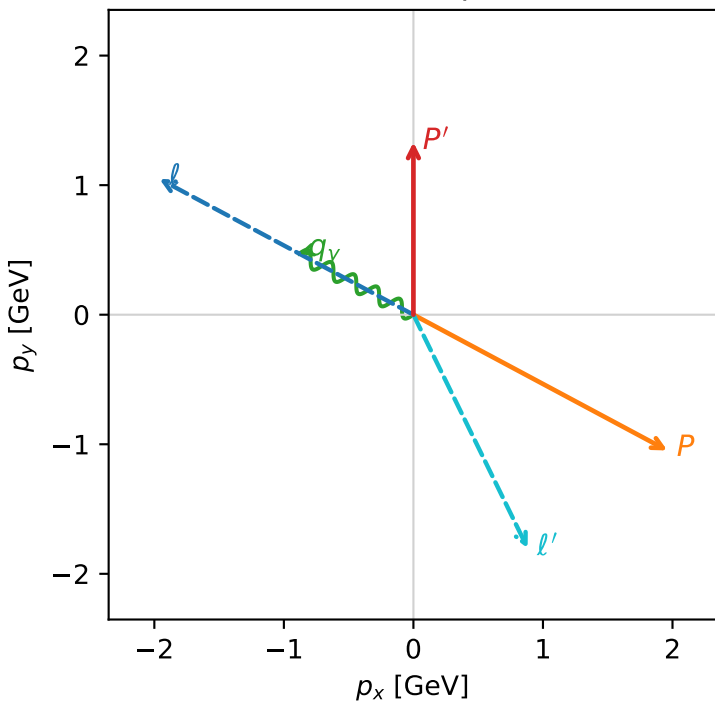


f3_mid_egamma_ll_region_3 F_3=0.0365473
 region: mid_Egamma_LL_region_3
 outgoing-state purity: 1
 kinematic point: high_s_high_theta_in_mid_Egamma
 incoming state: $|L_\mu\rangle \otimes |L_p\rangle$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.33148$
 $\theta_{in}=1.57079$, $\phi_l=2.65072$, $\phi_p=5.79231$
 $E_\gamma=1$, $\phi_\gamma=2.65072$
 $Q^2=16.1607$, $x_B=0.889769$, $t=-8.89083$, $W^2=2.88195$, $y=0.88063$
 $\theta(l', \gamma)=144.192$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 l $E= 2.2256$ $p_x= -1.9606$ $p_y= 1.0480$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 1.9606$ $p_y= -1.0480$ $p_z= 0.0000$
 l' $E= 2.0098$ $p_x= 0.8819$ $p_y= -1.8029$ $p_z= 0.0000$
 P' $E= 1.6287$ $p_x= 0.0000$ $p_y= 1.3315$ $p_z= 0.0000$
 q_γ $E= 1.0000$ $p_x= -0.8819$ $p_y= 0.4714$ $p_z= 0.0000$

Transverse plane



$h_e=-1, h_p=+1, h_\gamma=+1$
 muon L, proton L

$h_e=-1, h_p=-1, h_\gamma=+1$
 muon L, proton L

Leading initial-ensemble/final-state components (N=2, retained=97.8%)

